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Turning Analytics into Timely Action

*Caribbean Experience: Climate Information to
Decision Support for Anticipatory Action*

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Outline

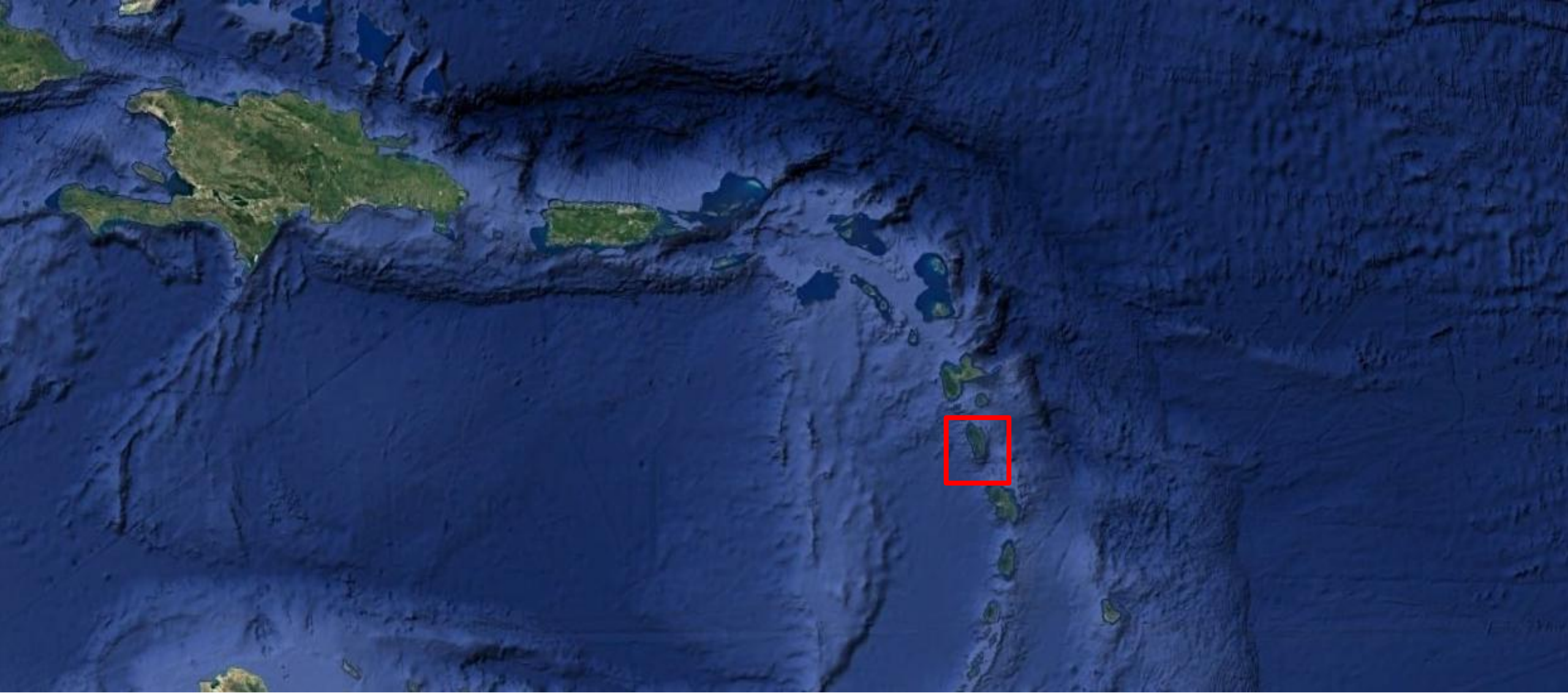
1. Context: Caribbean SIDS and the gap between climate information and action
2. What we set out to Do
3. Interventions: Precision irrigation (AWS-based) Pathogen detection (drone + machine learning)
4. Application: Translating data into real-time advisories
5. Link to Anticipatory Action: Acting before impacts occur



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Dominica





Dominica

■ Small Island Reality

- High climate variability
- Inefficient irrigation practices
- Limited real-time data for farmers and ministries

Key Gap: Climate information exists, but not translated into actionable farm decisions





What We Set Out to Do

Develop a
decision support
system for:

- Precision irrigation
- Disease risk monitoring



Integrate:

- Weather data
- Field-level observations
- Drone imagery and geospatial data



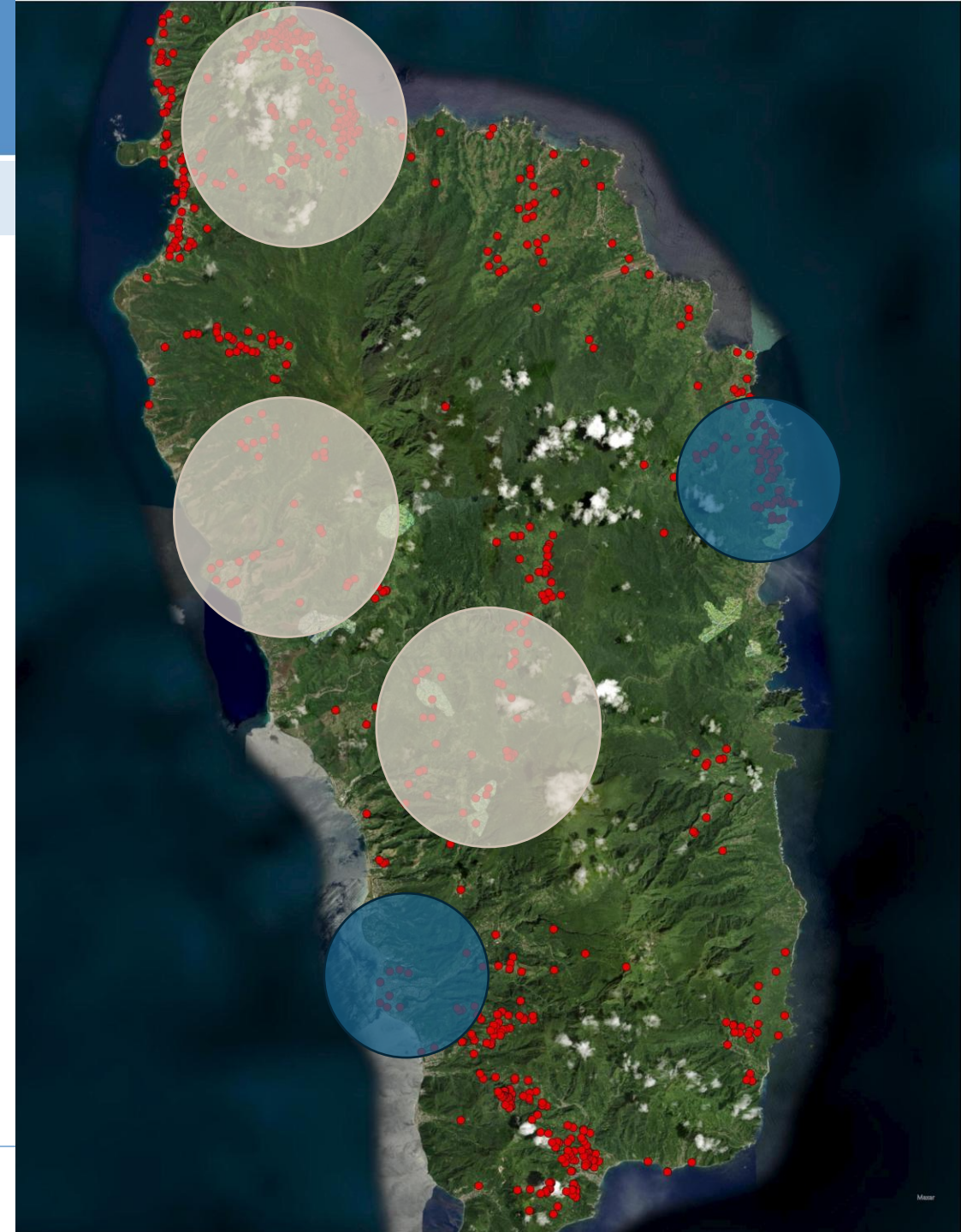
Deliver:

- Simple, actionable advisories for farmers and extension officers



Climate-Informed Precision Irrigation

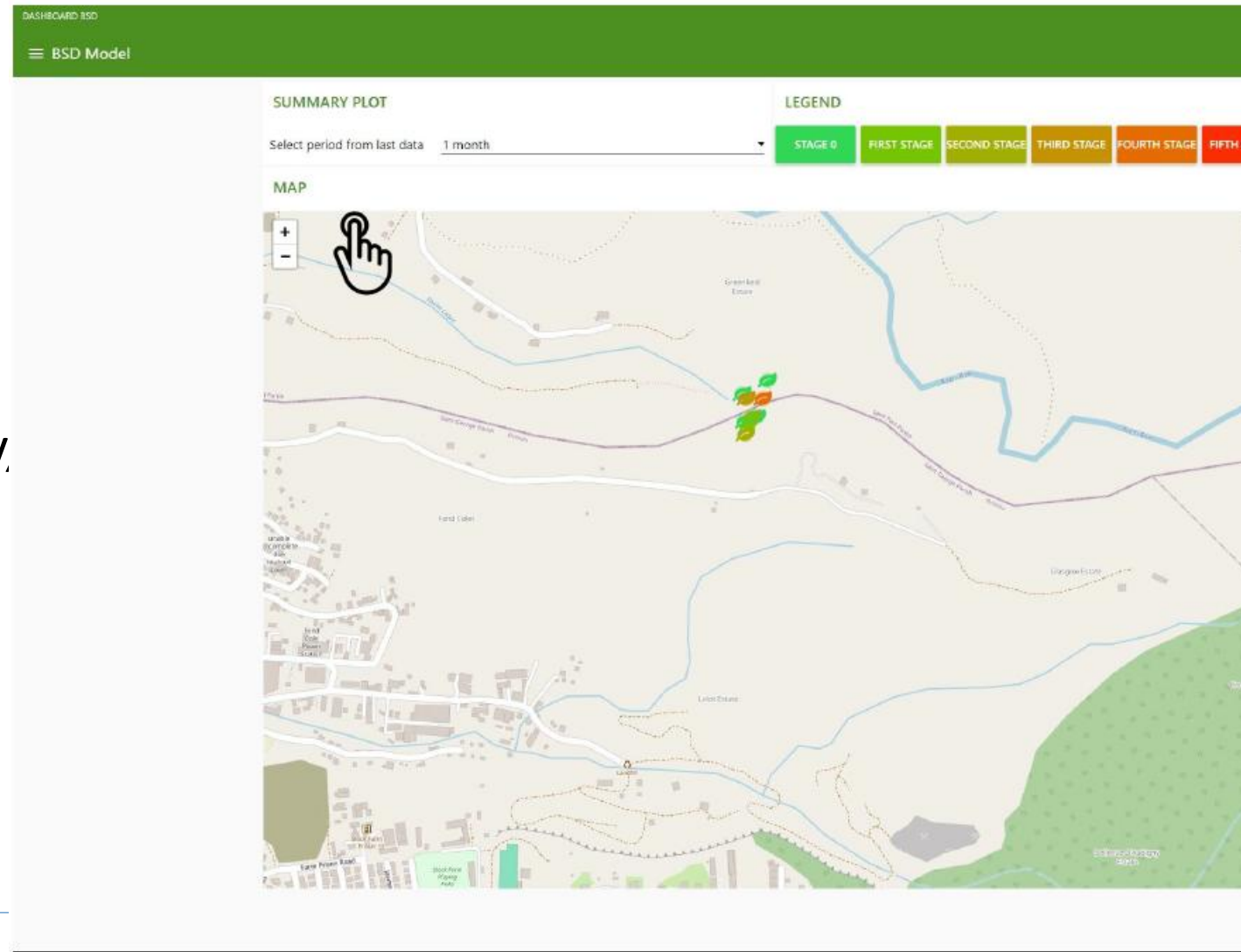
- Installation of **Automatic Weather Stations (AWS)**
- Collection of real-time weather data:
 - Rainfall
 - Temperature
 - Humidity
 - Soil moisture -3 depths
- Use of data to estimate:
 - Crop water requirements (based on crop type and field capacity)
- **Output:**
 - Irrigation advisories:
 - When to irrigate
 - How much to irrigate
 - When to delay irrigation





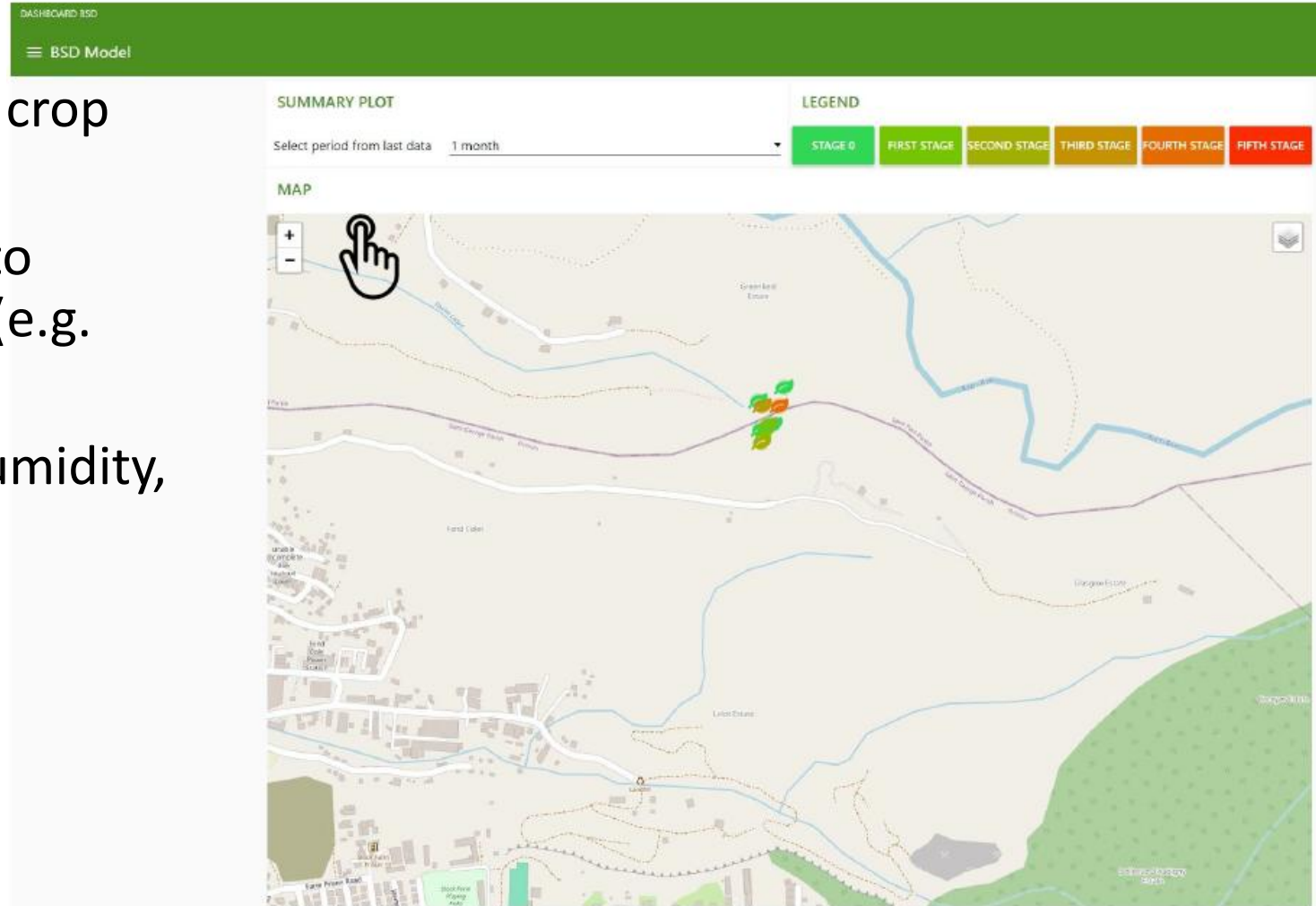
Pathogen Detection and Risk Monitoring

- Use of **drone imagery** to monitor crop conditions
- Application of **machine learning** to detect patterns linked to disease (e.g. BSD)
- Integration with weather data (humidity, rainfall)
- **Output:**
 - Early warning of disease risk
 - Targeted intervention guidance



Pathogen Detection and Risk Monitoring

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Linking to Anticipatory Action

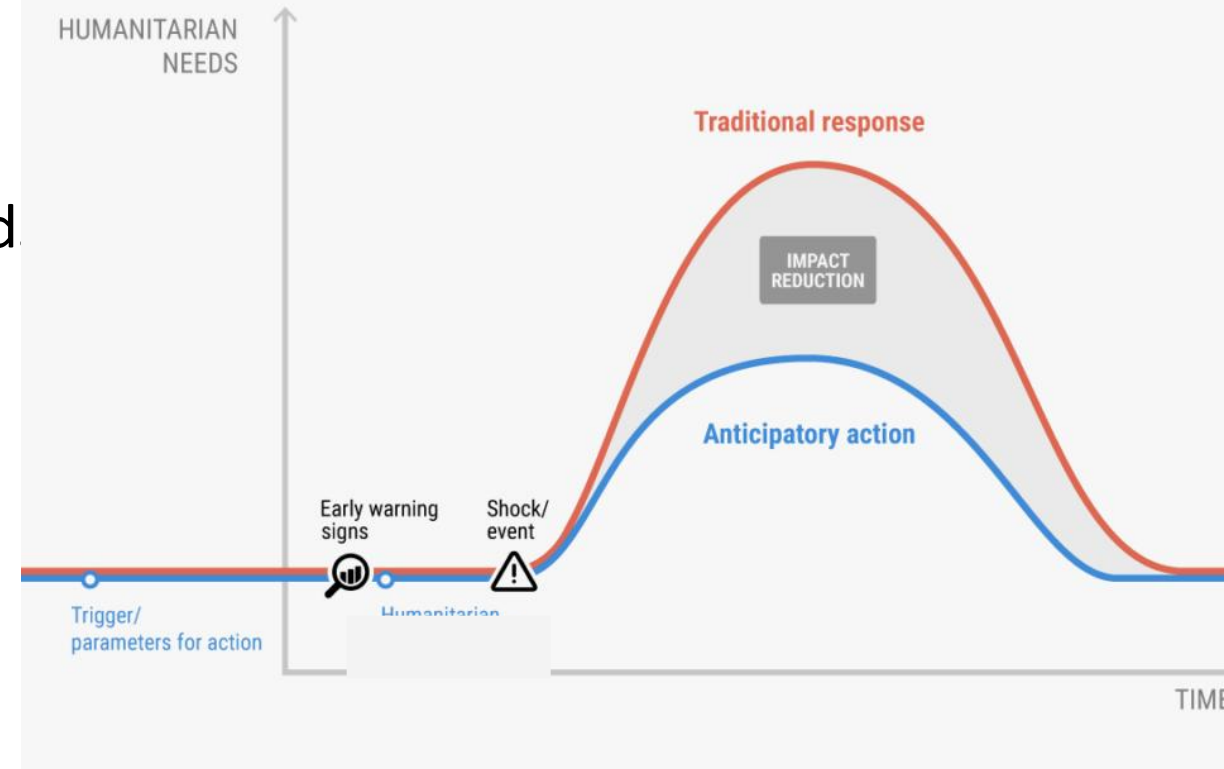
Anticipatory Action in Practice

- Act before water stress affects crops
- Enable early intervention before disease spread
- Use forecasts combined with real-time data

Result:

- Reduced crop losses
- More efficient use of water and inputs
- Improved targeting of interventions

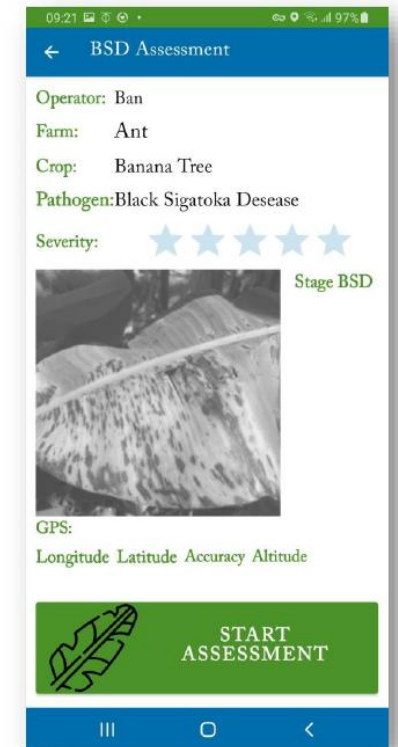
Decisions are made before impacts are visible in the field



Beyond Meteorological Triggers

Meteorological data is
necessary, but not sufficient

- Combines climate, farm-level, and spatial data
- Supports precision irrigation and pathogen detection
- Moves toward impact-based decision-making





From Climate Information to Anticipatory Action

Effective Decision Support

- Decision support links climate and field data to action
- Integration across data sources is critical
 - Systems must be usable and trusted by farmers and extension services

Why it matters:

- Enables action before impacts are visible
- Approach applicable across Caribbean SIDS
- Adaptable to multiple crops with minor calibration



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Thank You

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